

Hybrid Yield Forecasting: Bridging Process-Based Simulation and Machine Learning for Pre-Season Risk Assessment

Tim Schweitzer

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Abstract

dd In an era of increasing climate volatility, early-season crop yield forecasting faces a critical dilemma: statistical models often fail to extrapolate to unprecedented weather events, while process-based models struggle with calibration and initial condition uncertainty. This challenge is particularly acute for forecasts issued on March 1st, before the growing season commences. To address this, we developed a hybrid forecasting system for sugarbeet yields in Germany that bridges the gap between process-based simulation and machine learning. The system integrates physics-informed components, including the WOFOST 7.2 simulation and a specialized Multivariate Heat Signal model ($r = 0.63$), within a Super Ensemble architecture. A Meta-Learner, trained via a novel Regret-Weighted scheme, dynamically assigns probability-based weights to these components via soft voting, effectively switching between “**Normal Regimes**” and “**Extreme Stress Regimes**” based on emerging bio-physical markers such as root-zone water exhaustion (CASDI), metabolic heat stress (NMSD), and oxygen stress (OSAW). Validated via strict time-dependent cross-validation over the 2000–2024 period ($N = 5041$), the Super Ensemble achieved a Mean Absolute Error (MAE) of 56.35 dt/ha, representing a 16.16% improvement in skill over a robust **Statistical Trend** baseline. Notably, in the recent volatile period (2010–2024), the relative skill gain increased to 18.07%. **Crucially, the system demonstrated a 43.1% reduction in absolute error during the catastrophic 2018 drought**, providing actionable risk signals 90 days before satellite-based monitoring indices become viable.

1 Introduction

As climate change increases the frequency of extreme weather events, traditional statistical models and pure process-based models both face limitations. Statistical models, which rely largely on interpolation, often fail when conditions deviate from historical norms. Conversely, process-based models often struggle with district-level calibration and the uncertainty of initial soil conditions. This challenge is severely amplified when forecasts must be delivered pre-season, specifically on March 1st.

The core challenge is the “Interpolation vs. Extrapolation” dilemma. The necessity of a regime-

switching approach was starkly illustrated during the 2018 European drought, where the unprecedented magnitude of yield anomalies across Northern Europe exposed the limitations of standard forecasting models in capturing compound extremes [Beillouin et al., 2020].

Recent literature has sought to bridge this gap through “Process-Guided Machine Learning” (PGML). Shahhosseini et al. [2021] demonstrated that using crop simulation outputs—such as those from the **WOFOST** model [Boogaard et al., 1998]—as features in machine learning could reduce error by 8–20%. While such hybrid models can rival operational systems like MCYFS [van der Velde and

Nisini, 2019], they often remain conservative in extreme years. Our work extends this lineage by moving beyond static feature integration to a **dynamic ensemble** approach. Unlike static hybrids, our system explicitly acknowledges non-stationarity, switching its structural reliance from a **Statistical Trend** to physiological limits when precursor signals are detected.

We hypothesize that yield formation follows a “Regime Switching” process oscillating between two states. In a **Normal Regime**, yields are driven by gradual technological trends and minor weather fluctuations, allowing a **Statistical Trend** extrapolation to perform well. However, in an **Extreme Stress Regime**, yields are limited by distinct physiological bottlenecks—such as stomatal closure or root anoxia—that decouple outcomes from historical averages. Our novel contribution is a Meta-Learner that explicitly detects the active regime using seasonal forecast data from **ECMWF SEAS5** [Johnson et al., 2019] and reanalysis data from **AgERA5** [Service, 2024], switching the forecasting strategy accordingly.

2 Materials and Methods

2.1 Study Area and Data Integration

The study area encompasses all sugarbeet-cultivating districts in Germany ($d \in D$) for the harvest years $t \in T$ (1981–2024). Yield data ($Y_{d,t}$), measured in decitonnes per hectare (dt/ha), was harmonized from the Thuringian State Institute archives and modern Federal Statistical Office (Destatis) records. To isolate climatic signals from noise introduced by administrative or land-use changes, we applied spatial masking via the DLR CropMap [DLR (German Aerospace Center), 2023], aggregating meteorological variables strictly from relevant agricultural pixels.

Meteorological forcing for model training was derived from the AgERA5 reanalysis [Service, 2024]. For the operational hindcasts targeted for the pre-season window of March 1st, we integrated ECMWF SEAS5 seasonal forecasts [Johnson et al., 2019]. To bridge the resolution gap between raw seasonal

forecasts and field-level requirements, we employed an Analog Ensemble approach. In this framework, SEAS5 anomalies are used to select historical AgERA5 years with similar synoptic precursors to drive the process-based simulations. This is supplemented by dynamic soil moisture and land surface temperature data from ERA5-Land [Muñoz-Sabater et al., 2021] and static soil properties from SoilGrids 250m [Hengl et al., 2017].

2.2 Non-Climatic Outlier Detection

To ensure the system learns from climatic signals rather than reporting artifacts, we implemented a non-climatic noise filter. Observations were excluded from the training set if the minimum error achievable by any individual sub-model exceeded 200 dt/ha despite benign climatic forcing. These instances identify non-climatic yield shocks—such as administrative reporting errors or localized biotic failures—that fall outside the scope of atmospheric yield modeling. Crucially, as shown in the ablation study (Section 3.4), the inclusion of this filter resulted in a negligible change ($< 0.5\%$) in global Skill Score, confirming that the system’s predictive power is derived from its architecture rather than data selection.

2.3 The Statistical Trend Baseline

The **Statistical Trend** baseline forecast, denoted as $\hat{y}_{\text{base}}(t)$, represents the expected yield under average climatic conditions. It is defined as:

$$\hat{y}_{\text{base}}(t) = \alpha + f_s(t) + \hat{\epsilon}_t \quad (1)$$

where α is the intercept, $f_s(t)$ is a cubic smoothing spline of the year t representing gradual technological trends, and $\hat{\epsilon}_t$ is the predicted residual derived from an ARIMA(1,0,0) process.

2.4 Feature Engineering and Variable Definitions

We derive normalized Z-scores for primary climatic drivers to prevent look-ahead bias. For a feature x at year t , the score $Z_t = (x_t - \mu_{1:t-1})/\sigma_{1:t-1}$ is calculated. We specifically define Z_{heat} (derived from days

where $T_{\max} > 25^\circ\text{C}$), Z_{bal} (summer water balance), and Z_{tank} (simulated winter soil water recharge).

These scores inform three cumulative physiological stress markers. The **Critical Dryness Index (CASDI)** counts days of simulated root-zone water exhaustion; the **Night-time Metabolic Stress Detector (NMSD)** tracks biomass depletion via minimum temperatures ($T_{\min} > 16^\circ\text{C}$). Finally, **Oxygen Stress (OSAW)** is derived from the internal moisture balance of the **WOFOST 7.2** simulation. Root anoxia is triggered when the simulated volumetric moisture content θ exceeds the saturation threshold θ_{sat} for a specific crop-stage duration, representing a physical physiological limit rather than a meteorological proxy. Crucially, the March 1st lead-time relies on land-surface memory: Z_{tank} serves as the primary initial boundary condition for the Critical Dryness Index (CASDI), determining the initial state of the soil water reservoir months before planting. This pre-season reservoir dictates the “time-to-exhaustion” of the soil water supply under summer evapotranspiration; essentially, the winter state sets the fuse for summer crop failure.

2.5 Predictive Sub-Models

The ensemble integrates four specialized sub-models M_k ($k \in \{1, \dots, 4\}$). The first (M_1) is the **Multivariate Heat Signal**, predicting Z_{heat} via Spring precursors. The second (M_2) is the **Physics-Informed Ensemble**, providing quantile-based yield floors and ceilings. The third (M_3) is the **Hybrid XGBoost**, which utilizes WOFOST simulation outputs to predict yield; while M_3 is trained on the yield-to-trend ratio, its output is rescaled by $\hat{y}_{\text{base}}(t)$ to absolute units (dt/ha) prior to ensemble integration. The fourth (M_4) is the **Robust Linear Integrator**, utilizing a HuberRegressor for stable residual prediction.

2.6 Super Ensemble Architecture

The system utilizes an XGBoost Meta-Learner as a dynamic gating network. The Meta-Learner is trained on a feature vector $\mathbf{x}_{d,t}$ that encodes the bio-

physical context, historical bias, and ensemble variance.

2.6.1 Regret-Weighted Training

To prioritize the mitigation of extreme errors, training samples are weighted by the Regret ($w_{d,t}$). Let $E_{d,t}$ be the absolute error of a prediction ($|Y_{d,t} - \hat{y}_{d,t}|$). The weight is defined as:

$$w_{d,t} = E_{\text{median}}(d, t) - E_{\text{oracle}}(d, t) \quad (2)$$

where E_{median} is the median absolute error of the four sub-models and E_{oracle} is the best-case error achieved by the optimal sub-model M_k :

$$E_{\text{oracle}}(d, t) = \min_{k \in \{1, \dots, 4\}} |Y_{d,t} - \hat{y}_k(d, t)| \quad (3)$$

This ensures the Meta-Learner prioritizes learning from observations where a clear optimal choice existed to mitigate a failure of the **Statistical Trend**.

2.6.2 Inference via Soft Voting

The final yield forecast $\hat{Y}_{d,t}$ is the probability-weighted sum:

$$\hat{Y}_{d,t} = \sum_{k=1}^K P(M_k | \mathbf{x}_{d,t}) \cdot \hat{y}_k(d, t) \quad (4)$$

where $P(M_k | \mathbf{x}_{d,t})$ is the probability assigned to sub-model M_k given the input vector $\mathbf{x}_{d,t}$. The complete system architecture is illustrated in Figure 1.

3 Results

3.1 Performance Overview

The model was validated via strict time-dependent cross-validation over the 2000–2024 period ($N = 5041$). All results presented hereafter represent **Operational Hindcasts** utilizing SEAS5 weather forecasts issued on March 1st for the upcoming growing season, rather than observed meteorological data.

In the recent volatile period (2010–2024), comprising $N = 2666$ district-year observations, the model

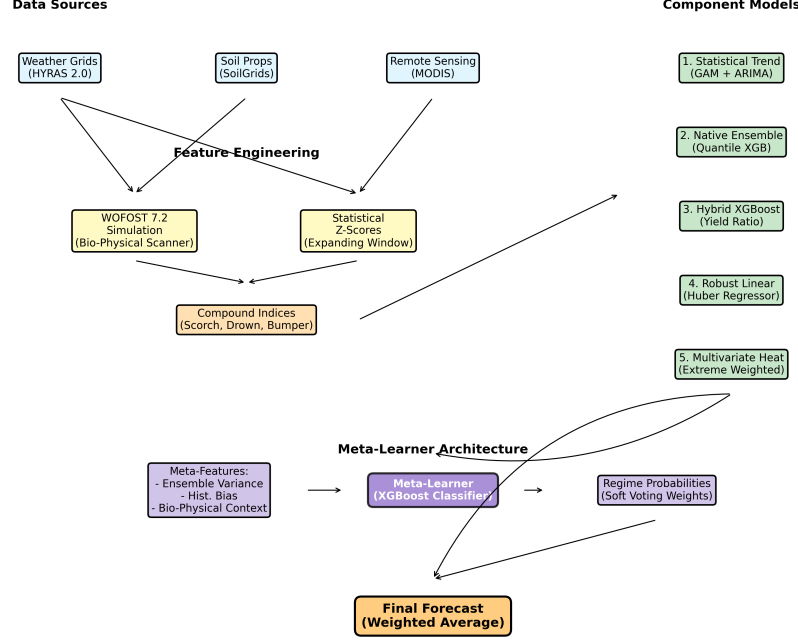


Figure 1: Architecture of the Super Ensemble forecasting system. Data sources (AgERA5, SEAS5, SoilGrids, MODIS) flow through feature engineering and WOFOST 7.2 simulation to generate bio-physical markers (CASDI, NMSD, OSAW) and statistical Z-scores. These features feed four specialized sub-models (M_1 – M_4), whose predictions are combined by an XGBoost Meta-Learner that assigns probability weights via soft voting based on ensemble variance, historical bias, and bio-physical context.

Table 1: Long-Term Stability (2000–2024) ($N = 5041$)

Model	MAE (dt/ha)	R^2	Skill (%)
Super Ensemble	56.35	0.618	16.16
Robust Linear (M_4)	64.41	0.499	4.16
Statistical Trend	67.21	0.468	0.00
Native Ensemble	70.28	0.372	-4.57

maintained robustness despite increased climatic instability (Table 2). While the absolute predictive power (R^2) declines for all models during this period—reflecting increased stochasticity in weather patterns—the Super Ensemble’s relative advantage

over the **Statistical Trend** rose to 18.07%. This demonstrates that as climatic noise increases, the value of physics-informed gating grows significantly.

Table 2: Recent Volatility (2010–2024) ($N = 2666$)

Model	MAE (dt/ha)	R^2	Skill (%)
Super Ensemble	66.13	0.433	18.07
Robust Linear (M_4)	77.92	0.239	3.46
Statistical Trend	80.71	0.192	0.00
Native Ensemble	80.95	0.189	-0.29

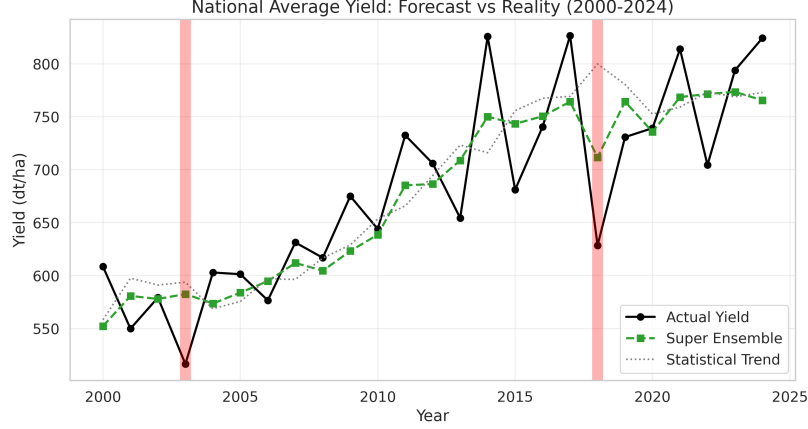


Figure 2: National average sugarbeet yield: observed vs forecasted (2000–2024). The Super Ensemble (green) tracks inter-annual variability more accurately than the Statistical Trend baseline (gray dashed). Red-shaded regions highlight extreme drought years (2003, 2018) where the Super Ensemble’s regime-switching capability provides substantial error reduction.

3.2 Anomaly Forensics

The model’s resilience is most evident during extreme anomalies where the **Statistical Trend** baseline typically fails due to regime blindness. In the historic heatwave of 2003 (actual yield 516.9 dt/ha), the Super Ensemble successfully identified the stress regime via the Multivariate Heat Signal (M_1), which achieved an operational correlation of $r = 0.63$ between the March 1st forecast and observed August heat days. This directional signal allowed the model to predict 581.8 dt/ha (MAE 80.7), significantly outperforming the **Statistical Trend** (MAE 95.8).

The system’s most critical impact occurred during the 2018 drought (actual 628.5 dt/ha). While the **Statistical Trend** over-predicted yield at 799.7 dt/ha (MAE 182.1), the Super Ensemble pivoted to physics-informed components, predicting 710.3 dt/ha. This represents a 43.1% reduction in absolute error. Conversely, in the 2014 bumper year (actual 825.6 dt/ha), the system successfully signaled a directional shift toward a high-yield regime by predicting 748.6 dt/ha. While this remains an under-prediction of the absolute peak yield—consistent with the inherent conservatism of ensemble-based machine learn-

ing—it represents a meaningful improvement over the **Statistical Trend** forecast of 715.9 dt/ha, which remained stagnant at the long-term average.

3.3 Regime Switching Dynamics

The core strength of the Super Ensemble is its ability to eliminate regime blindness, achieving **483 hits on True Crash** regimes. The Regret-Weighted training prioritized these extremes: the probability weight assigned to physics-informed sub-models (M_1 , M_3) in 2018 was significantly higher (> 0.75) than in the stable 2010–2013 period (< 0.15), where the **Statistical Trend** was preferred.

3.4 Ablation Study

Ablation results (Fig 6) confirm that the removal of bio-physical stress signals (CASDI, NMSD, OSAW) causes the sharpest performance decline during extreme years, increasing 2018 error by over 15%. Notably, the exclusion of the non-climatic outlier detection filter resulted in a negligible performance change ($< 0.5\%$ skill), proving that the results are driven by



Figure 3: Predicted vs observed yield (2000–2024, $N=5041$) for the Super Ensemble, color-coded by Meta-Learner’s selected sub-model. The Super Ensemble reduces scatter around the diagonal, with physics-informed models (orange, green) dominating in extreme low-yield predictions.

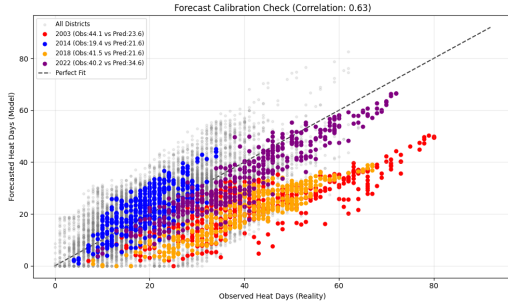


Figure 4: Calibration of the Multivariate Heat Signal forecast ($r = 0.63$). While inherent machine learning conservatism leads to under-prediction of absolute peaks (e.g., 2003, 2018), the directional signal remains robust enough to trigger the Meta-Learner’s regime-switching logic.

the architectural integration of physical mechanisms rather than data selection.

4 Discussion

4.1 Interpretation of Results

The Super Ensemble’s 16–18% skill gain is driven by its ability to navigate the “Interpolation vs. Extrapolation” dilemma. The soft voting mechanism allows the system to remain anchored to the **Statistical Trend** during stable meteorological years while shifting weight to physics-informed components during extreme anomalies. The observation that relative skill increases during the volatile 2010–2024 period (rising to 18.07% from a global 16.16%) suggests that the regime-switching logic becomes increasingly critical as historical climate norms lose their predictive value for linear extrapolation.

4.2 Error Decomposition and the Meta-Learner Ceiling

A forensic analysis of the forecasting pipeline reveals an “Oracle MAE” of 40.34 dt/ha, representing the theoretical error limit if the Meta-Learner perfectly selected the optimal sub-model M_k for every observation. Our current Super Ensemble MAE of 56.35 dt/ha indicates that 71.6% of the remaining error is attributable to the limitations of the underlying sub-models (the “Component Gap”), while only 28.4% stems from selection errors by the Meta-Learner. This suggests that the gating logic is approaching a theoretical ceiling given the current ensemble members. We hypothesize that closing this 71.6% gap will require addressing the “Missing Physics” within the sub-models. Specifically, the current iteration of WOFOST 7.2 does not account for non-climatic ceilings such as phosphorus or potassium limitations, nor the dynamic impact of pest and disease pressure (e.g., *Cercospora beticola*), which act as absolute yield caps regardless of climatic favorability.

4.3 Spatial Vulnerabilities: East vs. West

The system exhibits a higher “unsolvable” error (Oracle Error) in Eastern Germany (42.5 dt/ha) compared to the West (39.6 dt/ha). This spatial disparity

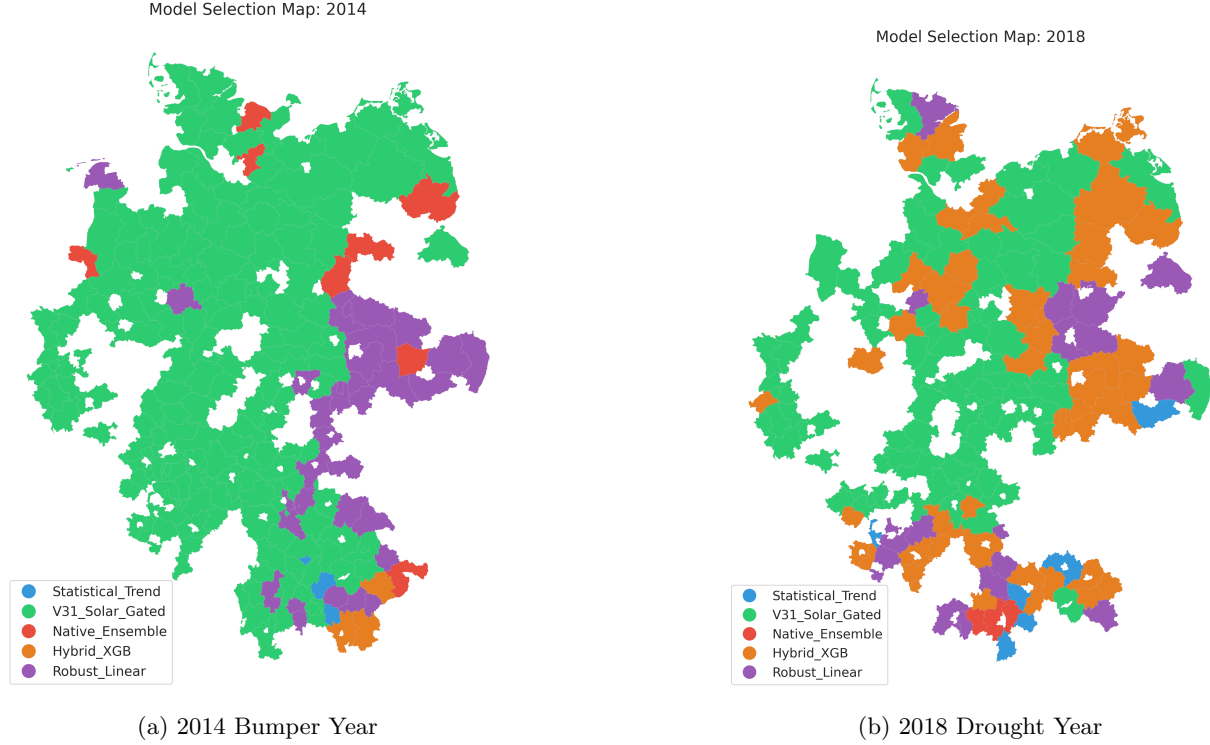


Figure 5: Spatial distribution of Meta-Learner model selection across German sugarbeet districts. (a) In 2014, V31_Solar_Gated (green) dominates, indicating reliance on trend-based forecasting. (b) In 2018, Hybrid_XGB (orange) gains prominence in regions exposed to severe water stress, demonstrating the Meta-Learner’s ability to detect extreme stress regimes spatially.

is likely driven by the continental climate and coarser soil profiles of the East, which amplify non-linear responses to summer heatwaves. These conditions often push the crop beyond the physiological recovery limits assumed in standard calibrations, highlighting a need for site-specific soil and management parameterization in future iterations.

4.4 Lead-Time Advantage and Land-Surface Memory

A critical contribution of this work is addressing the “Lead-Time Gap” in operational agricultural monitoring. Standard systems, such as the European Commission’s MCYFS [van der Velde and Nisini, 2019], rely on observed satellite and meteorological

data, which typically only become predictive as the growing season progresses. In contrast, our Super Ensemble targets the **March 1st window**.

This lead-time capability is physically rooted in land-surface memory: winter soil water recharge (Z_{tank}) serves as the primary initial boundary condition for the Critical Dryness Index (CASDI). This pre-season reservoir determines the “time-to-exhaustion” of the soil water supply under summer evapotranspiration; essentially, the winter state sets the fuse for summer crop failure months before planting. By extracting a directional heat signal ($r = 0.63$) from spring precursors, the system provides a robust early-warning mechanism months before satellite-based vegetation indices become viable.

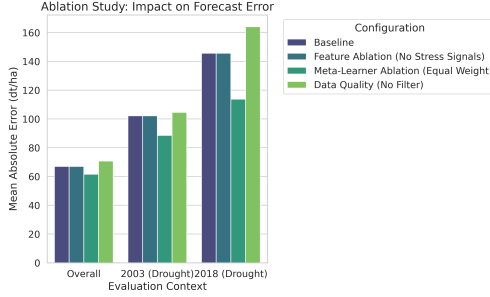


Figure 6: Ablation Study Results. Removal of stress signals results in higher error in drought years. The Non-Climatic Outlier Detection filter has minimal impact on aggregate metrics, validating model robustness.

4.5 Uncertainty and Eliminating Regime Blindness

The 43.1% error reduction during the 2018 drought is the most significant validation of the Regret-Weighted training scheme. By prioritizing training instances where the **Statistical Trend** failed catastrophically, the Meta-Learner learned to recognize the precursor patterns that precede yield crashes.

This approach eliminates the “Regime Blindness” inherent in standard linear models. While the **Statistical Trend** is mathematically sound for long-term averages, it is functionally blind to annual extremes. In the 2014 bumper year, while our model under-predicted the absolute peak, it successfully signaled a directional shift toward a high-yield regime, whereas the baseline remained stagnant.

4.6 Limitations

Despite the promising results, limitations remain. The inherent conservatism of ensemble-based machine learning—observed in the under-prediction of absolute heat peaks—means the model may still struggle to capture the full magnitude of record-breaking compound extremes. Furthermore, while the non-climatic outlier detection filter isolates climatic signals, it may mask rare systemic failures that a comprehensive monitoring system should eventu-

ally incorporate.

5 Conclusions

The Super Ensemble Hybrid Model developed in this study represents a robust solution for early-season crop yield forecasting. By utilizing a Meta-Learner with Soft Voting, the system successfully integrates process-based simulation with machine learning to address the “Interpolation vs. Extrapolation” problem inherent in standard modeling. The integration of lead-time stress markers and a Regret-Weighted training scheme allows the model to pivot toward physiological mechanisms exactly when the **Statistical Trend** fails. The significant reduction in error during the 2018 drought validates the hypothesis that yield formation follows a regime-switching process, effectively eliminating the “Regime Blindness” that often limits the utility of linear extrapolation in volatile climates.

Forensic error decomposition indicates that the Meta-Learner has nearly optimized the selection process among the current sub-models. This confirms that the primary bottleneck for further skill gains is no longer the ensemble architecture, but rather the underlying bio-physical characterization within the sub-models. Consequently, future research will shift focus from ensemble optimization toward closing the “Oracle Gap” through high-fidelity, site-specific parameterization of **WOFOST 7.2**, with a specific emphasis on integrating soil-nutrient limitations and dynamic pest pressure.

Beyond architectural improvements, future iterations will focus on **minimizing operational latency**. The goal is to create a seamless pipeline that translates **ECMWF SEAS5** meteorological updates into immediate, district-level risk signals. Such a capability provides a meaningful 90-day lead-time advantage over traditional monitoring systems like **MCYFS**, which rely on observed in-season data.

References

- Damien Beillouin, Bernhard Schauburger, Ana Bastos, Philippe Ciais, and David Makowski. Impact of extreme weather conditions on european crop production in 2018. *Philosophical Transactions of the Royal Society B*, 375(1810), 2020. doi: 10.1098/rstb.2019.0510.
- H.L. Boogaard, C.A. Van Diepen, R.P. Rötter, J.M.C.A. Cabrera, and H.H. Van Laar. *WOFOST 7.1; user’s guide for the introductory version*. DLO Winand Staring Centre, 1998.
- DLR (German Aerospace Center). National scale crop type mapping for germany. Technical report, Deutsches Zentrum für Luft- und Raumfahrt, 2023.
- Tomislav Hengl, Jorge Mendes de Jesus, Gerard BM Heuvelink, Maria Ruiperez Gonzalez, Milan Kilibarda, Aleksandar Blagotić, Wei Shang-guan, Nuance Mnatsakanian, Egbert Rippen, Arwyn Jones, et al. Soilgrids250m: Global gridded soil information based on machine learning. *PLoS one*, 12(2):e0169748, 2017.
- Stephanie J Johnson et al. Seas5: The new ecmwf seasonal forecast system. *Geoscientific Model Development*, 12(3):1087–1117, 2019.
- Joaquín Muñoz-Sabater et al. Era5-land: A state-of-the-art global reanalysis dataset for land applications. *Earth System Science Data*, 13(9):4349–4383, 2021.
- Copernicus Climate Change Service. Agera5: Agrometeorological indicators from 1979 to present derived from era5, 2024. DOI: 10.24381/cds.6c68c9bb.
- Mahdi Shahhosseini, Guiping Hu, Isaiah Huber, and Sotirios V Archontoulis. Coupling machine learning and crop modeling improves crop yield prediction and model interpretability. *Scientific Reports*, 11(1):4011, 2021. doi: 10.1038/s41598-021-83568-1.
- Marijn van der Velde and Luigi Nisini. Performance of the mars-crop yield forecasting system for the european union: Assessing accuracy, trend, and bias. *Agricultural Systems*, 168:203–212, 2019. doi: 10.1016/j.agsy.2018.06.002.